

CS-412 Operating Systems (Minor Test I)

Date: March 12, 2024

Time: 1 hr

Marks: 20

Answer all the questions. Each question carries 5 marks.

1. It is sometimes difficult to achieve a layered approach if two components of the operating system are dependent on each other? Identify and explain a scenario in which it is unclear how to layer two system components that require tight coupling of their functionalities.
2. Describe the actions taken by a kernel to context-switch between processes?
3. Discuss how the following pairs of scheduling criteria conflict in certain settings.
 - (a) CPU utilization and response time
 - (b) I/O device utilization and CPU utilization
4. Describe two kernel data structures in which race conditions are possible. Be sure to include a description of how a race condition can occur.

**School of Computer and Systems Sciences
Jawaharlal Nehru University, New Delhi**

**End Semester Examination, Winter Semester 2024
CS 412--Operating Systems**

Time: 3 Hrs. Date: May 17, 2024 Marks: 60

Note: Answer all the 6 questions. Each question carries 10 marks.

1. What is the main advantage of micro-kernel approach to system design? How it is different from layered operating systems.
2. What is the purpose of the system call **fork ()** in the UNIX operating system? Write a C routine to create a child process using the fork system call. Incorporate an error check in your routine in case the creation of the child process fails.
3. The implementation of mutex lock suffers from busy waiting. Describe what changes would be necessary so that a process waiting to acquire a mutex lock would be blocked and placed into a waiting queue until the lock become available.
4. What are the methods of handling deadlocks in operating systems? How we can prevent the deadlocks in operating systems?
5. What is pure demand paging. Explain the difference between following page replacement algorithms.
 - LRU replacement
 - FIFO replacement
 - Optimal page replacement
6. Explain the difference between external and internal fragmentation. How we can resolve the fragmentation issue.

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1. It is sometimes difficult to achieve a layered approach if two components of the operating system are dependent on each other? Identify and explain a scenario in which it is unclear how to layer two system components that require tight coupling of their functionalities.
2. Describe the actions taken by a kernel to context-switch between processes?
3. Consider the following five processes to run in a single CPU using preemptive Shortest Job First scheduling. Show the scheduling order of the processes using Gantt chart. What is the waiting time for each process?

Process	Arrival Time	Burst Time
P1	2	6
P2	5	2
P3	1	8
P4	0	3
P5	4	4

4. Consider how to implement a mutex lock. Assume that the following structure defining the mutex lock is available:

```
typedef struct {  
    int available;
```

```
} lock;
```

(available == 0) indicates that the lock is available, and a value of 1 indicates that the lock is unavailable. Using this struct, illustrate how the following functions can be implemented:

```
void acquire (lock *mutex)
```

```
void release (lock *mutex)
```

Be sure to include any initialization that may be necessary.

int available
mutex
mutex = 0
int --

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Time: 3 Hrs.

Date: May 17, 2024

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Note: Answer all the questions. Each question carry (6+6=12) marks.

1. (a) What is the main advantage of micro-kernel approach to system design? How do user programs and system services interact in a microkernel approach?

(b) What is the purpose of the system call **fork ()** in the UNIX operating system? Write a C routine to create a child process using the fork system call. Incorporate an error check in your routine in case the creation of the child process fails.

2. (a) A possibility for solving the Dining Philosopher problem is to impose a linear ordering on the semaphores. Assume that each philosopher process run its own thread, how this order can be imposed. Demonstrate the function in C language.

(b) Consider the following snapshot of a system:

	Allocation	Max	Available
	A B C	A B C	A B C
P0	1 1 2	4 3 3	2 1 0
P1	2 1 2	3 2 2	
P2	4 0 1	9 0 2	
P3	0 2 0	7 5 3	
P4	1 1 2	1 1 2	

Illustrate that the system is in a safe state by demonstrating an order in which the process may complete. Determine the total sum of each type of resource?

3. (a) This diagram shows an example of memory configuration under dynamic partitioning, after a number of placement and swapping-out operations have been carried out. Addresses go from left to right; gray areas indicate blocks occupied by processes; white areas indicate free memory blocks. The last process placed is 2 Mbytes and is marked with an X. Only one process was swapped out after that.

	4M		1M	X	5M		8M		2M		4M		3M	
--	----	--	----	---	----	--	----	--	----	--	----	--	----	--

- a. What was the maximum size of the swapped-out process?
- b. What was the size of the free block just before it was partitioned by X?
- c. A new 3-Mbyte allocation request must be satisfied next. Indicate the intervals of memory where a partition will be created for the new process under the following three placement algorithms: best-fit, first-fit and worst-fit. For each algorithm, draw a horizontal segment under the memory strip and label it clearly.
- (b) What is the purpose of paging the page tables? Describe all the steps taken by the Operating System in translating a logical address into a physical address.
4. (a) Consider a system that uses pure demand paging. When a process first start execution, how you characterize the page fault rate.
Consider the following page reference string:
6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0
Assuming demand paging with three frames, how many page faults would occur for the following page fault replacement algorithms?
- LRU replacement
 - FIFO replacement
 - Optimal page replacement
- (b) Compare the performance of C-SCAN, SCAN and FCFS disk scheduling, assuming a uniform distribution of requests. Consider the average response time, the variation in response time and the effective bandwidth.
5. (a) How does the principle of least privilege enhances the protection of Operating System? How can systems that implement the principle of least privilege still have protection failures that leads to security violations. Explain with an example.
- (b) Define the security in context of operating system. Confidentiality, integrity, and availability are three components of security. Describe an application, with example, that integrity and availability but not confidentiality, an application that requires confidentiality and integrity but not (high) availability, and an application that requires confidentiality, integrity, and availability.